

AHMED BOURMECHE

ahmed.bourmeche.eng@gmail.com — +216 28 013 411 — [GitHub](#) — [LinkedIn](#) — [Portfolio](#)

Education

Engineering Degree in Industrial IT & Automation — 4th Year 2022 – Present
National Institute of Applied Sciences and Technology (INSAT), University of Carthage
Relevant coursework: Real-Time Systems, Embedded Systems Design, Industrial Control, Signal Processing, Industrial Networks, Robotics

Experience

CTO & Co-Founder — CyberNexus 2024 – Present
cybernexus.tn

- Architected **ELTACHKILA**, a full-scale football community platform (**Python, TypeScript, MySQL, Docker + Nginx**): real-time WebSocket chat, stadium bookings, match scheduling, and secure JWT auth; sole DevOps owner (VPS, CI/CD, SSL).
- Built a complete **gamification engine** — XP system, missions, leaderboards, redeemable rewards store — and led a 4-person engineering team from design to production deployment.

Embedded Systems Intern — LAB619 (Industrial IoT Solutions) Jul 2025 – Sep 2025

- Developed a production-grade **Industrial IoT Node** on **ESP32-S3**: multi-protocol firmware (NB-IoT, RS485/232, CAN, LoRaWAN) for real-time industrial data acquisition and remote monitoring.
- Designed a **multilayer PCB** in KiCad with EMC-compliant layout; delivered validated prototype with full manufacturing files (Gerber, BOM, assembly).

Lead Developer — Lions Club Tunis INSAT (STEP'ACT 2026) Jul 2025 – Present
stepact.lionsinsat.tn

- Built and deployed a marathon registration platform (**React, TypeScript, Node.js, Docker**) handling real-time participant data and dynamic multi-distance pricing for a large-scale charity event.

Technical Projects

Autonomous Mobile Robot (Capstone Project) *Embedded C, STM32, KiCad, SolidWorks*
Designed a full autonomous robot on **STM32 Blue Pill**: finite state machine (Move/Detect/Stop/Rotate), redundant obstacle detection via fused **ToF + ultrasonic** sensors (front & rear), DC motor (N20) control with **PWM/PID**, and low-level STM32 peripherals (GPIO, timers, UART, interrupts). Modeled chassis in **SolidWorks**, produced a custom PCB for motor driving and sensor interfacing. Extended with an **ESP32 WiFi** module streaming live telemetry (x, y, θ) to an IoT dashboard.

IIoT Predictive Maintenance Dashboard — Github Repository *JavaScript, Python, ML*
Real-time sensor telemetry platform with **ML-based failure scoring**, configurable alert thresholds, anomaly timelines, and equipment health drill-down views targeting industrial predictive maintenance.

Quarter Car Suspension System *LabVIEW, Proteus, ATmega328P, C*
Hardware-in-the-loop simulation coupling **Proteus** (ATmega328P Euler integrator, $dt=10^{-4}s$) and **LabVIEW** via virtual COM for real-time 2-DOF suspension modelling. Proved non-linearity (-33% superposition error); implemented **PD controller** achieving -87% peak displacement suppression.

PID Control Dashboard — Github Repository *Python, React, Flask, MATLAB Simulink*
React dashboard live-coupled to a **MATLAB Simulink** digital twin via Flask REST/WebSocket; real-time PID tuning with $<100ms$ latency, disturbance injection, and telemetry CSV export.

LinkedIn Optimizer — Github Repository *JavaScript, Chrome Extension API*
Chrome extension (Manifest V3) injecting an **AI writing assistant** into LinkedIn's composer via shadow DOM isolation; real-time tone analysis and clarity scoring with zero external data collection.

Hand Gesture & Voice Control Systems — Github Repository *Python, OpenCV, MediaPipe*
Gesture-controlled maze game (MediaPipe + Tkinter) and hands-free YouTube controller via computer vision and browser speech recognition — multi-modal HCI prototypes focused on accessibility.

Leadership

Treasurer — Lions Club Tunis INSAT Jul 2025 – Present
Managing club finances and project budgeting for charitable initiatives including STEP'ACT 2026, a large-scale fundraising marathon.

Project Manager — IEEE RAS 2023
Led development of a computer-vision-based autonomous line-follower robot; coordinated hardware and firmware sub-teams.

Skills

Languages	C, Embedded C, Python, TypeScript, JavaScript, SQL
Embedded & Hardware	STM32, ESP32, ATmega328P, KiCad (multilayer PCB)
Protocols	LoRaWAN, CAN, RS485/232, UART, SPI, I2C
Software & Tools	React, Node.js, Flask, Docker, MySQL, WebSocket, Git, CI/CD
Engineering Tools	MATLAB Simulink, LabVIEW, Proteus, KiCad, SolidWorks